BIONEXUS STUDIO AI

Complete Platform Architecture, Features & Audit Report

Date: 9/21/2026 | Platform Executive Summary

1. PRODUCT VISION & WORKFLOW INTEGRATION

- **Core Vision:** BioNexus Studio AI transforms scattered web bioinformatics tools into a connected, beginner-friendly scientific workspace.
- **Connected Workflow:** NCBI / FASTA -> Sequence Validation -> Sequence Analysis -> Primer Design -> BLAST Specificity -> MSA & Phylogenetics -> AI Explanation -> PDF Report.
- **Live Site:** https://bionexus-studio-ai.netlify.app

2. CORE FEATURES & MODULE SUMMARY

- **NCBI Explorer:** Integrated Entrez E-utilities API to search Nucleotide, Protein, Gene, and PubMed databases; fetch FASTA sequences and summaries directly into analysis pipelines.
- **Primer Designer:** Nearest-Neighbor Tm calculation (SantaLucia 1998), GC%, length range checks, self-dimer & cross-dimer analysis, top 5 ranked primer pairs with quality score.
- **BLAST Center:** BLASTN, BLASTP, BLASTX, TBLASTN, TBLASTX alignment engine with interactive SVG Hit Alignment Map (Query vs Hit overview) and E-value thresholds.
- **Sequence Analysis:** DNA reverse complement, GC%, 6-frame translation, ORF Finder; Protein molecular weight, Isoelectric Point (pI) prediction, and amino acid composition.
- **Alignment Hub (MSA):** Multiple Sequence Alignment using progressive Needleman-Wunsch algorithm with Clustal color mapping, conservation scores, and consensus sequence.
- **Phylogenetics Engine:** Neighbor-Joining (NJ) & UPGMA tree reconstruction; Jukes-Cantor (JC69), Kimura 2-Parameter (K80), and p-Distance models; SVG tree renderer & Newick (.nwk) exporter.
- **Protein Structure:** Interactive 3D molecular viewer powered by 3Dmol.js supporting Cartoon, Stick, Sphere, and Surface rendering for PDB entries (e.g., 1TUP).
- **AI Bioinformatics Assistant:** Context-aware AI assistant with "Audit My Page Work" feature that inspects current sequence state and parameter choices to give expert recommendations.

3. PHASE 1 DNA & FASTA VALIDATION ENGINE

- **Validation Engine:** Built real-time DNA & FASTA validator (sequenceValidator.js) detecting canonical bases (A,T,G,C,U), IUPAC ambiguity codes, whitespace, and non-DNA characters.
- **Protein Input Guard:** Detects protein sequence inputs pasted into DNA primer fields and offers a 1-click option to fetch cDNA directly from NCBI.
- **ValidationPanel UI:** Reusable live feedback component showing length (bp), GC%, AT%, ambiguous base count, and explicit invalid character lists.

4. 10-POINT AUDIT & SCIENTIFIC VERIFICATION (PASS)

- **Does It Work?:** PASS - All 9 application tabs execute expected workflows smoothly without freezing or crashing.
- **Scientific Accuracy:** PASS - Nearest-neighbor Tm, codon translation, Jukes-Cantor distance, and BLAST scoring match standard literature.
- **Invalid Input Rejection:** PASS - Non-DNA bases, negative range bounds, and protein sequences are rejected before execution.
- **UI Error Clarity:** PASS - Clear amber/red alert cards explicitly state validation failures and provide remediation steps.
- **Loading & API Failure:** PASS - Animated loading spinners display during API calls; network errors are caught gracefully in try/catch blocks.

- **Mobile Responsiveness:** PASS - Responsive CSS flex/grid layouts with max-width 768px breaking tabs into accessible mobile cards.

5. PHYLOGENETICS MODULE DEBUGGING & FIXES

- **Algorithms Fixed:** Implemented Neighbor-Joining (NJ) Saitou & Nei 1987 algorithm; fixed UPGMA matrix lookup bounds bug; added Jukes-Cantor (JC69), Kimura 2-P (K80), and p-Distance models.
- **Test Suite:** Verified across 10 test cases including exact 4-species test data, unequal length alignment, invalid characters, duplicate sequence headers, and highly divergent sequences.